



Software Engineering

Introduction

Software Development Life Cycle (SDLC)

- ❖ SDLC is the processes and methodologies used in creating software.
- ❖ There are different SDLCs; but all must include 5 fundamental activities:
 - Software specification (requirement analysis)
 - Software Design
 - Software implementation
 - Software validation (testing)
 - Software evolution (maintenance and further development)
- ❖ IEEE provides standards to document all the processes within SDLC.
- ❖ Some [popular SDLC models](#) are:
 - Waterfall, V-Model, Incremental, Iterative and Agile

IEEE software life cycle

SQA – Software quality assurance IEEE 730

SCM – Software configuration management IEEE 828

STD – Software test documentation IEEE 829

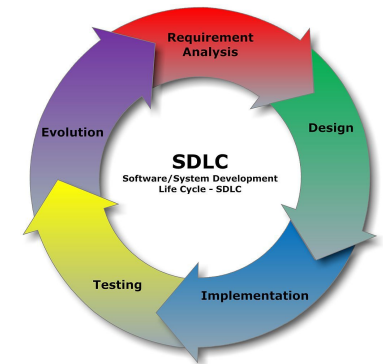
SRS – **Software requirements specification IEEE 830**

V&V – Software verification and validation IEEE 1012

SDD – Software design description IEEE 1016

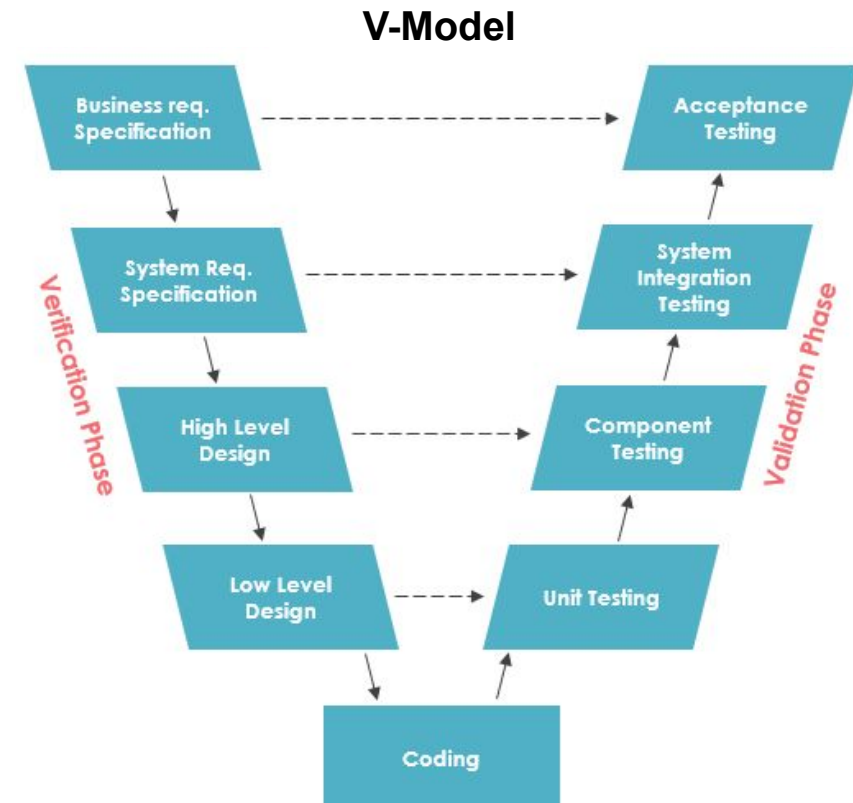
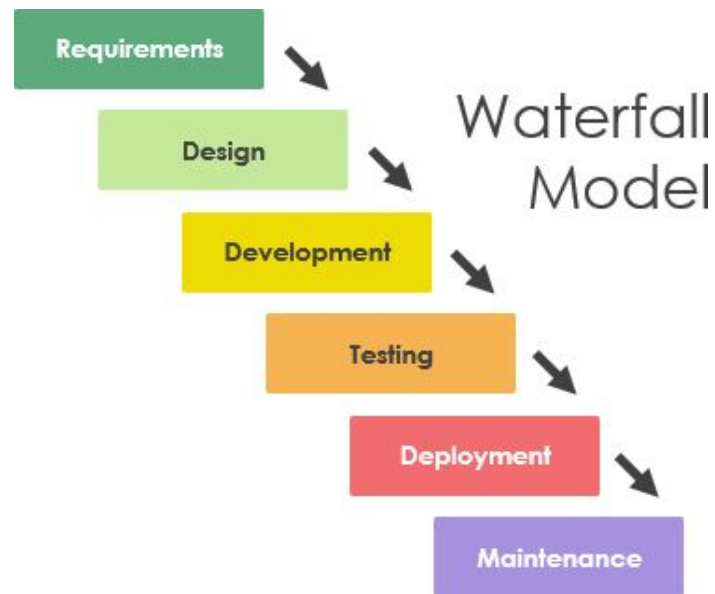
SPM – Software project management IEEE 1058

SUD – Software user documentation IEEE 1063



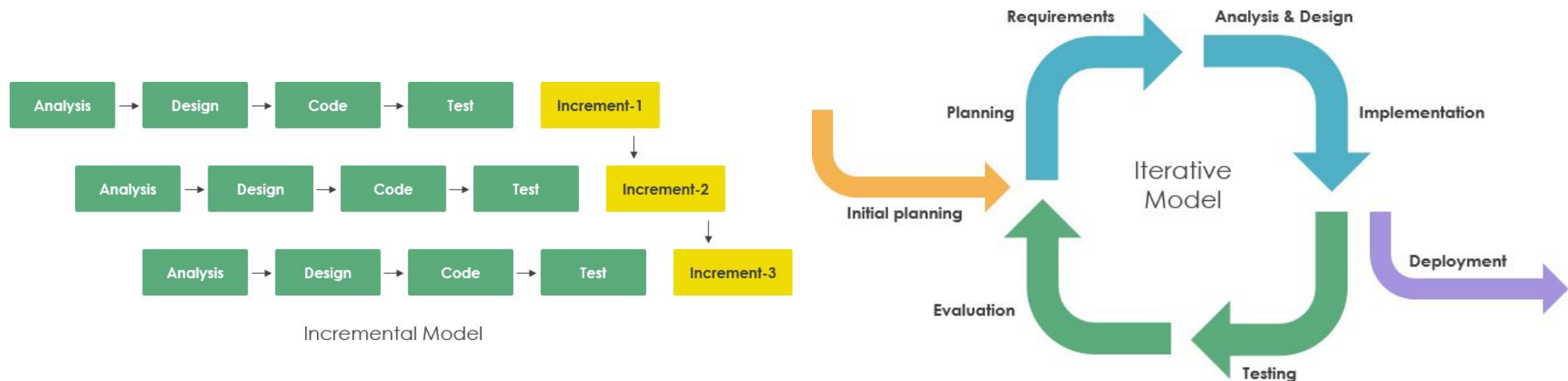
SDLC - Waterfall and V-Model

- ❖ In **Waterfall**, the whole process is divided into sequential phases and each phase needs to be completed before the next starts.
- ❖ **V-Model** is an extension of the waterfall model.
 - It is known as the **V**erification and **V**alidation Model



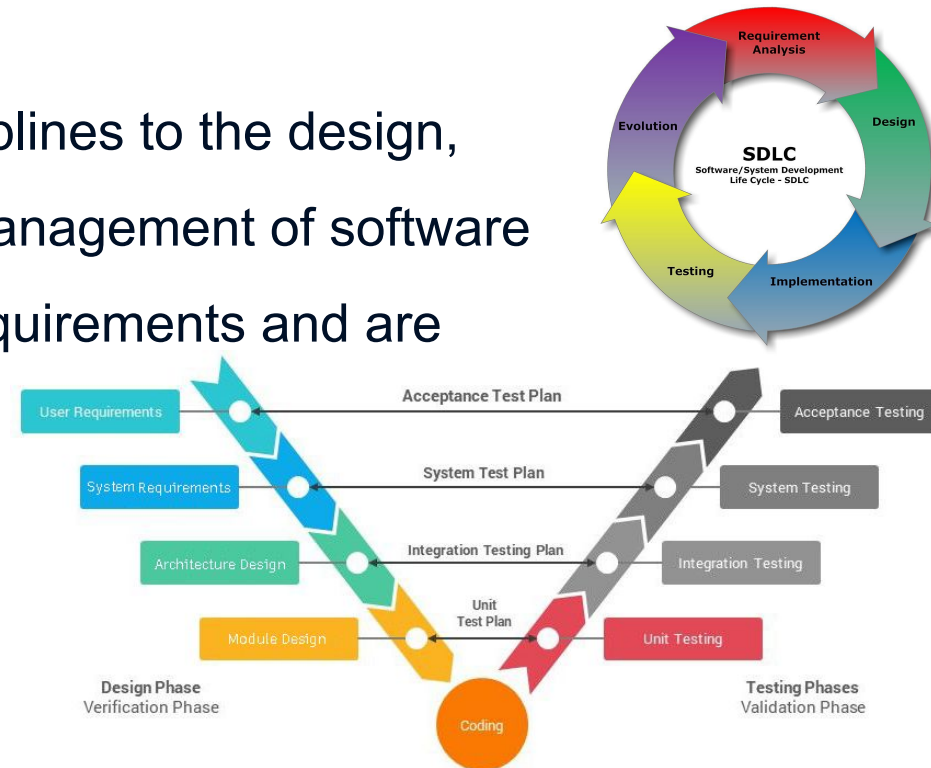
SDLC - Iterative and Incremental Models

- ❖ In the **iterative** model, development does not start with a full specification of requirements. it is suitable for scenarios where the final solution is hard to be defined.
- ❖ In the **incremental** model a software is designed, implemented and tested incrementally until the product is finished. This model combines waterfall model with the iterative philosophy.
- ❖ Both provide the development team with feedback from the previous increment or iteration.



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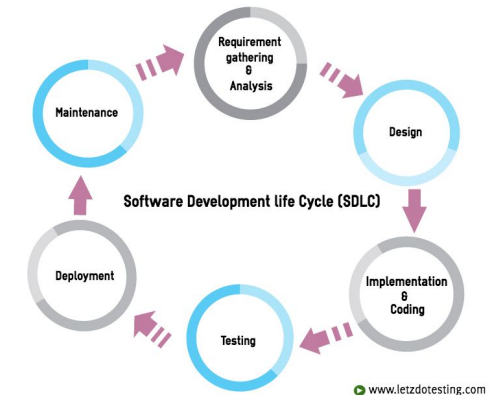
- ❖ *Software engineering* is a systematic and disciplined approach to the development, operation, and maintenance of a software.
- ❖ It involves engineering principles and disciplines to the design, development, testing, deployment, and management of software systems to ensure they meet specified requirements and are reliable, scalable, and maintainable.
- ❖ It is an *evolving* science, and there are lots of techniques and practical knowledge
 - Different types of system require different techniques



Software Engineering - Key Aspects

❖ **Software Development Lifecycle (SDLC):**

- *Requirement Analysis*: Understanding what the software needs to do.
- *Design*: Structuring the software's architecture and components.
- *Implementation(Coding)*: Writing the actual software code.
- *Testing*: Ensuring the software works as intended.
- *Deployment*: Making the software available for use.
- *Maintenance*: Updating and fixing software after deployment.



- ❖ **Principles of Engineering**: Software engineers apply engineering concepts like modularity, abstraction, and optimization to make efficient, reliable, and easy to maintain softwares.
- ❖ **Tools and Techniques**: Using a variety of programming languages, frameworks, development environments, and tools to develop software. Common tools include version control systems (e.g., Git), IDEs, and testing frameworks.

Software Engineering - Key Aspects

- ❖ **Project Management:** Involving managing projects, timelines, and resources.
 - Using Agile, Scrum, Waterfall and ect. to organize and manage software projects.
- ❖ **Quality Assurance (QA):** Ensuring software meets quality standards through testing, debugging, and validation processes and it is a core aspect of software engineering.
- ❖ **Collaboration and Teamwork:** Often involves working in teams, with roles like developers, testers, designers, and project managers collaborating to deliver software.
- ❖ *Software engineering* allows us to *tackle* complex problems, *minimize* costs and *increase* quality, acceptability, reliability, security, safety, efficiency and maintainability
- ❖ The **goal** of software engineering is to produce high-quality software that is delivered on time and within budget, while meeting the needs of users and stakeholders.



Software Engineering

❖ Some useful links

- [Software Development Life Cycle Models](#)
- [What is a Software Process Model?](#)
- [Agile vs Waterfall](#)
- [What is Agile?](#)
- [Scrum Guide](#)
- [What is Scrum?](#)
- [What is Software Engineering?](#)
- [What is software engineering?](#)
- [An Introductory Guide to Software Engineering Basics and Concepts](#)
- [Introduction to Software Engineering](#)
- [Software Engineering](#)